

JEE MAIN REVISION MASTER: ALCOHOLS, PHENOLS AND ETHERS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 60 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Alcohols Phenols And Ethers - Core Application	60 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Alcohols Phenols And Ethers - Core Application (Very High Yield — 60 PYQs)

Core Principle: Key problem pattern in Alcohols Phenols And Ethers.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Consider the following reaction. Statement I : In the above reaction, product formed will be a mixture of benzyl alcohol and iodobe..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Identify the colour of compound 'X' in the sequence of the reaction...." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "A hydroxy compound (X) with molar mass 122gmol^{-1} is acetylated with acetic anhydride, using a large excess of the reagent ensuring co..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Identify the colour of compound 'X' in the sequence of the reaction.

- (A) Violet
- (B) Green
- (C) Red
- (D) Colourless

Correct Answer: None

Solution Analysis: Compound X is phenolphthalein (an acid–base indicator). In basic solution, phenolphthalein first shows a pink colour. But when NaOH is in excess (very strong base), phenolphthalein changes to a form that is colourless. So, the colour of compound X in excess NaOH is colourless .